

# Cheng-Han Lin

Full-stack developer · Taichung, Taiwan

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Computer science graduate and full-stack developer with hands-on experience building production LLM systems — retrieval-augmented generation, self-hosted model inference and machine-learning forecasting — alongside the web, mobile and payment infrastructure around them. I ship complete products solo and care about evaluation, privacy and the failure cases most people skip.

## Projects

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### HeartBox — AI mood-journaling platform (RAG + emotion forecasting)

Solo developer · Capstone project · [heartbox.tw](https://heartbox.tw) · [github.com/alanlin0604/HeartBox](https://github.com/alanlin0604/HeartBox)

Designed, built and deployed a mental-health platform end to end, combining retrieval-grounded LLM feedback with a machine-learning emotion forecast. Sole developer across ML, backend, frontend, mobile and infrastructure. **Live demo at [heartbox.tw](https://heartbox.tw) — sign in with test1 / test1, no registration required.**

- **RAG pipeline** — BGE-M3 embeddings over a ChromaDB vector store built from seven clinical sources (WHO, APA, NHS, NIMH). Retrieval triggers when an entry's sentiment falls below  $-0.4$  and returns the top-3 relevant passages, which the model must cite — so advice is traceable to a source rather than hallucinated.
- **Random Forest forecasting** — engineered 53 features (12 behavioural and physiological metrics  $\times$  four lag windows, plus five calendar features) over 22,796 training rows. 5-fold cross-validation: MAE 0.22 on sentiment (range  $-1$  to  $+1$ ), MAE 1.04 on the stress index (range 0–10), and AUC 0.948 with 88% recall on high-stress-day classification over 31,720 rows — deliberately tuned for recall, since a missed warning costs more than a false alarm.
- **Model selection** — evaluated linear regression, a single decision tree, Random Forest, XGBoost and an LSTM against data requirements, interpretability, inference cost and overfitting risk. Chose Random Forest for sub-50 ms CPU inference that trains on only a few hundred rows per user while staying interpretable through feature importance.
- **Self-hosted inference** — ran open-weight models behind a FastAPI GPU service exposed through a Cloudflare Tunnel: LLaVA-v1.6-Mistral-7B for vision, and TAIDE-LX-7B, an open-weight model tuned for Traditional Chinese. Journal text stays inside a controlled environment instead of going to a third-party API.

- **Privacy and safety** — journal contents encrypted with Fernet (AES-128-CBC + HMAC-SHA256) using environment-injected keys, plus 2FA. Three-step separated consent modelled on GDPR Art. 7 and Taiwan’s PDPA, with AI-training consent independently refusable and parental confirmation required for users aged 13–17. Crisis-keyword detection across journals, AI chat and the anonymous community surfaces national helplines immediately.
- Also delivered: PHQ-9 / GAD-7 assessments, mood trend and correlation visualisations, sleep and habit analytics, a 103-achievement system, and an Android build via Capacitor.

**LapseWatch — subscription renewal reminder service**

Solo developer · Jun 2026 – present · [lapsewatch.smallworks.app](#)

- Built a service that notifies users **before** subscriptions auto-renew, through LINE, email and desktop notifications, with Google Calendar sync and CSV / JSON export.
- Developed a Chrome extension that detects subscriptions automatically, reading only a narrow window of text around matched keywords to minimise data collection.
- Integrated recurring subscription billing with statutory e-invoicing (ECPay, Taiwan’s e-invoice system); designed privacy-first, with no bank account linking and no cross-site tracking.

**PantryKeeper — shared household inventory tracker**

Solo developer · Jun 2026 – present · [pantrykeeper.net](#)

- Built a multi-user inventory and expiry tracker with per-member permission controls, offline support and waste statistics.
- Wrote a bulk import parser that accepts pasted data from Notion, Excel and Google Sheets, inferring quantity and unit from free-form text.

**Skills**

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| Area      | Detail                                                                                                                                                                                                        |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages | Python, JavaScript, TypeScript, SQL, HTML, CSS                                                                                                                                                                |
| AI / ML   | Retrieval-augmented generation, LangChain, ChromaDB, embedding models (BGE-M3), self-hosted LLM inference, scikit-learn, Random Forest, feature engineering, cross-validation, model evaluation and selection |
| Backend   | Django, Django REST Framework, Django Channels (WebSocket), FastAPI, Node.js, REST API design, PostgreSQL                                                                                                     |

| Area           | Detail                                                                                             |
|----------------|----------------------------------------------------------------------------------------------------|
| Frontend       | React, Vite, Tailwind CSS, Recharts, Chrome extension development                                  |
| Infrastructure | Google Cloud Run, Cloudflare Pages and Tunnel, Capacitor (Android), Git, CI/CD                     |
| Security       | Fernet / AES encryption, JWT, 2FA, rate limiting, GDPR and Taiwan PDPA consent design              |
| Integrations   | Payment gateway and statutory e-invoicing (ECPay, Taiwan), LINE Messaging API, Google Calendar API |

## Education

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**National Chin-Yi University of Technology** — Taichung, Taiwan M.S. in Computer Science and Information Engineering · from September 2026

**National Chin-Yi University of Technology** — Taichung, Taiwan B.S. in Computer Science and Information Engineering · graduated June 2026

Capstone project: HeartBox (above) · Student Association, Administration Division

## Languages

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Mandarin Chinese (native) · English (professional reading and writing)

## Contact

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alan930604@gmail.com · github.com/alanlin0604 · linkedin.com/in/chenghanlin-tw